HOST-TRK: Home Oriented Smart Technologies-Training and Readiness Kit

C. Judilla , D. Bickram, C. Hawkins, Y. Li, H. Wang
Technology for Occupational Performance Lab, University of Florida

Problem Statement

Home-oriented smart technologies (HOST), also known as smart home technology (SHT), offer transformative potential for enhancing safety and autonomy for older adults and people with disabilities (PWD).[3,11] However, a significant "digital burden" prevents adoption: nearly 40% of users report usability challenges, and many experience a "fear of failure" when interacting with complex systems.[1,2,4,7,8] Current solutions often prioritize device installation over user education, leaving clients and occupational therapists (OTPs) without effective tools to practice essential skills. Without hands-on training, SHTs are frequently abandoned or misused, particularly in high-stakes situations such as falls or emergencies. [9]

Background

As populations age and the demand for aging-in-place solutions increases, HOST systems are frequently recommended as scalable interventions to support home safety and independence. [2,3,10,11] These technologies are often introduced during care transitions or post-installation, where users are expected to learn system behavior in real-world environments with little opportunity for structured practice. For individuals with cognitive, mobility, or sensory impairments, this learning curve can be particularly steep, amplifying anxiety and reducing trust in automation. [7,8]

Occupational therapy practitioners (OTPs) play a critical role in preparing clients for safe and independent technology use; however, existing clinical workflows lack standardized, portable tools for HOST skill development. [5] In the absence of training platforms that allow users to safely explore automation, simulate common disruptions, and build confidence through repetition, smart home systems fail to achieve their intended impact and are disproportionately abandoned by those most at risk. [9]

Methods/Approach/Solutions Considered

The design approach focused on enabling users to develop practical competence and confidence with HOST before deployment in real living environments. Rather than treating HOST adoption as a one-time installation task, this project framed technology use as a functional skill that must be practiced and reinforced across varied daily scenarios. Emphasis was placed on reducing cognitive load, supporting experiential learning, and enabling clinician-guided training in low-risk settings.

Several solution categories were considered. Software-based approaches, including mobile applications and video tutorials were evaluated for their scalability and ease of distribution. However, these approaches were limited in their ability to support hands-on learning, simulate real-world failures, or accommodate PWD. Screen-based instruction also assumes baseline digital literacy and confidence, which conflicts with the goal of reducing digital burden.

Wearable-centered safety systems and app-driven emergency response tools were also considered. While effective for detection and alerting, these systems typically emphasize

passive monitoring rather than user understanding, and they often reinforce stigma by requiring visible, on-body devices. In addition, wearables do not support skill-building related to environmental automation, troubleshooting, or system recovery, limiting their usefulness as training tools.

In-home, post-installation training was explored as another alternative. While contextually relevant, this approach provides limited opportunity to practice uncommon but critical scenarios (e.g., power loss, false alarms) and restricts training to a single home configuration. This limits both modularity and breadth of training scenarios, and it reduces the ability of clinicians to generalize instruction across clients.

These limitations highlighted the need for a portable, hands-on training platform that supports modular reconfiguration, accommodates a wide range of scenarios, and enables repeated practice without real-world consequences. A physical, smart system was therefore selected to prioritize ease of use, tactile interaction, and clinician-led instruction, while maintaining flexibility to represent diverse smart home configurations and failure conditions. This approach directly informed the development of the final training solution described in the following section.

Final Approach & Design

The Smart Home Technology Training Board is a portable, modular, and interactive educational tool designed to build user confidence and competence through experiential learning.

Key Design Features:

1) Modular "Plug-and-Play" Architecture: Utilizing a task board design with interchangeable modules, the system allows for the integration of various SHT devices such as smart locks, motion sensors, smart bulbs, and medication dispensers.
2) Scenario-Based Simulation: The board uses "Scenario Cards" to guide users through step-by-step simulations of real-life challenges.
    a. Examples: Responding to a nighttime fall, managing a power outage, or setting a medication reminder.
3) Neuro-Inclusive & Accessible Interfaces: The hardware supports multimodal interaction, including voice-command workflows (Amazon Alexa/Google Nest), high-contrast visual cues, and simplified physical buttons to accommodate users with various cognitive and physical impairments.
4) Portability for Diverse Settings: The kit is lightweight and battery-powered, ensuring it can be used in rural or resource-limited environments with inconsistent internet/power.

Technical Feasibility & Prototyping

The prototype leverages the Gator Smart House infrastructure for real-world validation. It builds upon our previous work developing "Activity of Daily Living (ADL)" boards, which have been successfully used by researchers at multiple universities to quantify assistive technology performance.

Cost Estimation:
Plywood is the primary material used for the main structure of the training board. The estimated total cost is approximately $106 per kit, making it an affordable option, especially for long-term use.

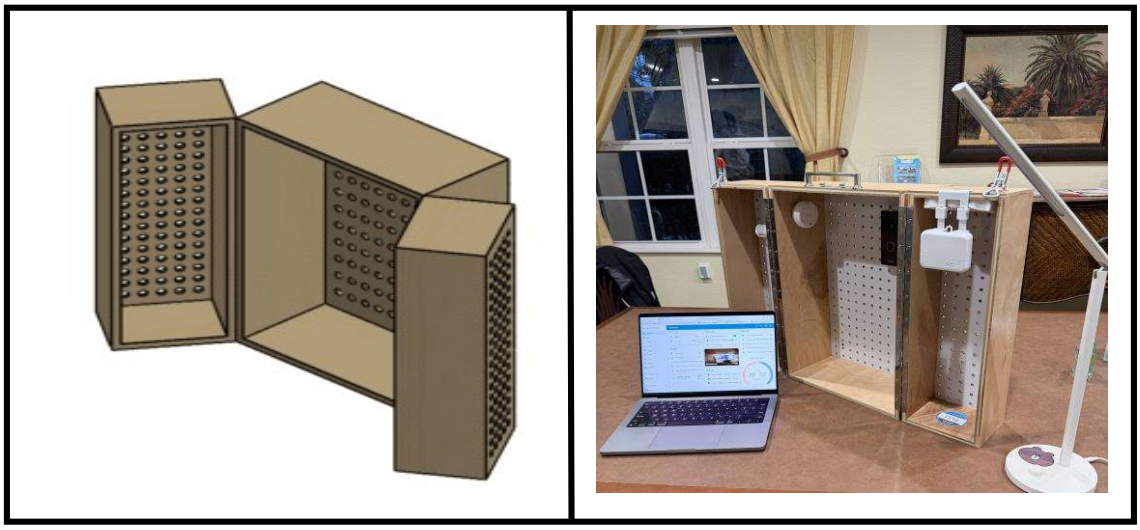

Figure 1. Isometric view of design (left) and prototype photo with HOST devices installed (right)

User-Centered Design & Research
The board's development is grounded in focus groups with 45 participants and interviews with service providers, using a human centered design to address key barriers to adoption.

1) Addressing the "Pride Factor": Research showed that many users resist "wearable" help-seeking devices due to stigma. The training board emphasizes Passive Environmental Safety (e.g., lights that activate automatically when a user stands up), which users find more acceptable and less intrusive.

2) Reducing Maintenance Fatigue: Users identified "troubleshooting" as a major barrier. The board includes specific modules to teach "Zero-Configuration" skills and auto-reconnection protocols.

Outcome & Potential Impact

Empowering Independence:
By enabling users to actively control their home environment rather than passively receive alerts, the system helps reduce preventable emergencies and supports safer, more independent living.

Clinical Integration:
The kit gives occupational therapists (OTPs) a standardized, hands-on tool to assess smart home technology readiness and deliver structured, reimbursable training that builds real-world functional skills.

Inclusivity:
The design prioritizes individuals with mobility or cognitive challenges, nearly 35% of adults, ensuring that those most at risk of losing independence can safely age in place.

Table 1. How the HOST-TRK will address the user needs and challenges

Current Challenges	Design Solutions: HOST Training Board	Outcome/Impact
"Fear of Failure" & Anxiety: Many users experience high anxiety when first using HOST for fear of breaking devices or causing false alarms.	**Risk-Free Simulation:** The board allows users to practice high-stakes scenarios (e.g., managing a simulated nighttime fall) in a safe, controlled environment, building "perceived safety confidence" before home deployment.	Builds confidence and reduces stress anxiety.
"Digital Burden" of Setup: Setup and troubleshooting are major barriers; 77% of seniors require help with initial configuration.	**Zero-Configuration Training:** The kit uses "plug-and-play" modules to teach users how to manage automatic reconnections and power-loss recovery without complex pairing.	Enables independent use without tech support.
Stigma & The "Pride Factor": Users often reject active wearables (like PERS buttons) because they signify a loss of independence or "decrepitude".	**Focus on Passive Actuation:** Training emphasizes invisible environmental sensors (e.g., motion-activated lighting) that improve safety without requiring active help-seeking or stigmatizing accessories.	Increases long-term adoption and dignity.
Limited Clinical Training Tools: Occupational therapists lack standardized tools to assess "HOST readiness" or provide hands-on patient education.	**Interdisciplinary Clinical Framework:** The board is designed as a standardized tool for OTPs to evaluate functional performance and integrate technology use into formal care plans.	Streamlines therapist training and reimbursement.
Accessibility Barriers: Many standard smart devices have complex interfaces that are unsuitable for people	**Multimodal Accessibility:** The board integrates voice-command workflows, high-contrast visuals, and simplified tactile buttons to support	Supports diverse cognitive/physical needs.

with cognitive or mobility impairments.	neuro-inclusive and physically accessible design.	
Internet & Power Dependence: Traditional HOST often fails in rural or resource-limited areas due to poor broadband or frequent power outages.	**Portable & Offline Functionality:** The kit features battery-powered, preconfigured devices that can operate offline, making it a viable educational tool in any environment.	Works reliably in any environment.

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